

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 35)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

[1 mark]

- A Elemental (liquid) mercury
- B Inorganic mercury compounds
- C Organic (methyl) mercury
- D Mercury vapour

☐
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0 1 . 2

Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

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0 1 . 3

Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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5

0 2 . 1

Explain the roles of **three** named groups of soil organisms in maintaining soil fertility.

[6 marks]

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0 2 . 2 Explain why a soil pH between 5.5 and 7.0 is best for fertility.

[2 marks]

0 2 . 3 Explain how soil compaction by heavy machinery reduces fertility.

[2 marks]

10

0 3 Table 1 shows some properties of four groups of insecticide.

Table 1

Insecticide group	Toxicity to insects	Toxicity to mammals	Persistence	Liposolubility
Organochlorines (eg DDT)	High	Low	High	High
Organophosphates (eg malathion)	High	High	Low	Low
Pyrethroids	High	Low	Low	Low
Neonicotinoids	High	Low	Moderate	Low (water soluble)

03 . 1

Use Table 1 to explain why organochlorine insecticides caused the decline of birds of prey such as the peregrine falcon in the 1950s and 1960s.

[4 marks]

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03 . 2

Organophosphates replaced organochlorines for many uses. Use Table 1 to give **one** advantage and **one** disadvantage of this change.

[2 marks]

Advantage

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Disadvantage

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03 . 3

Explain why pyrethroids should not be used near rivers or fish farms.

[1 mark]

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03 . 4

Describe how the properties of neonicotinoids allow them to be used as a seed coating, and explain why this method of application reduces harm to non-target species.

[3 marks]

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0 4 . 1

Describe how bioleaching is used to extract copper from low-grade ores or spoil heaps, and explain why it may be more sustainable than smelting.

[5 marks]

0 4 . 2

Describe the process of phytomining.

[3 marks]

0 4 . 3

Explain how polymer adsorption may provide a low-energy source of uranium in the future.

[2 marks]

10

0 5

Table 1 shows the mass of carbon in the main reservoirs of the carbon cycle.

Table 1

Reservoir	Mass of carbon / billion tonnes
Atmosphere (CO ₂)	700
Living organisms	470
Dead organic matter (soil)	3 700
Surface ocean waters	500
Deep ocean (dissolved)	35 000
Fossil fuels	10 000
Sedimentary carbonate rocks	20 000 000

0 5

1

Express the mass of carbon in the atmosphere in standard form, in tonnes.

[1 mark]

Answer

tonnes

0 5

2

Calculate the percentage increase in atmospheric carbon that would result from burning 10% of the fossil fuel reservoir, assuming all the carbon remained in the atmosphere.

[2 marks]

Answer

%

0 5 . 3

Explain why the sedimentary carbonate rocks, although by far the largest reservoir, have little effect on atmospheric CO₂ concentrations.

[2 marks]

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0 5 . 4

Describe how carbon moves from the atmosphere into sedimentary carbonate rocks.

[3 marks]

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0 5 . 5

Explain why aerobic respiration returns carbon to the atmosphere as CO₂ but anaerobic respiration returns it mainly as methane.

[2 marks]

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06 . 1

Explain how affluence and the strategies of manufacturers increase the amount of solid waste produced.

[4 marks]

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06 . 2

Give **three** factors that affect the choice of disposal method for a solid waste.

[3 marks]

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06 . 3

Explain **one** environmental advantage and **one** disadvantage of composting garden and food waste rather than sending it to landfill.

[2 marks]

Advantage

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Disadvantage

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06 . 4

Explain why mixed municipal waste is difficult to recycle.

[1 mark]

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07

Table 1 shows the energy used per passenger-kilometre by different forms of transport when carrying a typical load.

Table 1

Mode of transport	Typical occupancy	Energy use / MJ per passenger-km
Car (petrol, 1 occupant)	1	2.9
Car (petrol, 4 occupants)	4	0.7
Bus (diesel)	25	0.6
Electric train	200	0.3
Bicycle	1	0.06
Short-haul aircraft	120	2.4

07

1

Calculate the percentage reduction in energy per passenger-km achieved by carrying four people in a car instead of one.

[1 mark]

Answer

%

07

2

Explain why bulk transport (buses, trains) uses less energy per passenger-km than cars.

[2 marks]

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07

3

Suggest **three** reasons why people continue to use cars despite the energy advantages of public transport.

[3 marks]

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0 7 . 4 Explain how integrated transport systems and Active Traffic Management reduce energy use.

[4 marks]

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10

0 8 . 1 Explain why electronic meters must be calibrated before use and describe how a pH meter is calibrated.

[3 marks]

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08 . 2

Describe how wind velocity measurements with an anemometer should be standardised.

[3 marks]

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08 . 3

Describe how a whirling hygrometer is used to measure humidity.

[2 marks]

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08 . 4

Describe **two** methods of measuring nitrate concentration in water.

[2 marks]

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09

Table 1 shows features of three rechargeable battery types.

Table 1

Battery type	Energy density / Wh kg ⁻¹	Charge-dis charge cycles	Storage efficiency / %	Cost per kWh / £	Safety issues
Lead-acid	35	500	80	100	Toxic lead, acid
Nickel-cadmium	50	1500	75	250	Toxic cadmium
Lithium-ion	200	2000	92	150	Fire risk

09 . 1

Use Table 1 to explain why lithium-ion batteries are used in electric vehicles but lead-acid batteries are still used for stationary storage in some off-grid solar systems.

[4 marks]

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09 . 2

Give **two** other factors that affect the viability of a rechargeable battery.

[2 marks]

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09 . 3

Describe how Vehicle-to-Grid (V2G) systems could reduce the need for peaking power stations.

[2 marks]

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09 . 4

Explain how Power-to-Gas (P2G) systems use existing infrastructure.

[2 marks]

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Ozone depletion is often described as an environmental problem that has been tackled successfully. Compare the effectiveness of the methods used to restore the ozone layer with the effectiveness of attempts to tackle global climate change, and explain the reasons for the difference.

[9 marks]

Extra space

10 . **2** Name the international agreement of 1985 that preceded the Montreal Protocol and state its main weakness.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

OR

1 1 . 2

Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

END OF QUESTIONS